

Notes: Bessembinder, Hao, and Zheng (RFS, 2020)

Liquidity Provision Contracts and Market Quality: Evidence from the New York Stock Exchange

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1 Big picture

- **Question.** Do designated market maker contracts improve market quality, and do liquidity-provision incentives spill over to competing venues?
- **Setting.** The NYSE DMM contract strengthens obligations and rebates when prior-month consolidated average daily volume is below 1 million shares.
- **Identification.** The paper uses a regression discontinuity design around the 1-million-share threshold.
- **Main result.** Enhanced DMM obligations are associated with narrower quoted and effective spreads, greater depth, and more frequent price improvement.
- **Mechanism.** Much of the improvement occurs off the NYSE, supporting strategic complementarity in liquidity provision.
- **Policy message.** DMM contracts can improve market quality even in electronic limit order markets where any trader can provide liquidity.

2 Data and measurement

2.1 NYSE DMM threshold

NYSE Rule 104 defines active securities as those with consolidated average daily volume at least 1 million shares during the prior month. Less-active securities receive the enhanced DMM contract: the DMM must quote at the NBBO at least 15% of the day and receives a higher rebate; active securities face a 10% requirement.

Interpretation. The threshold creates quasi-random variation in DMM obligations for securities whose volume fluctuates around 1 million shares.

2.2 Sample and outcomes

The sample covers NYSE-listed nonfinancial and nonutility securities from September 2009 to December 2013. The baseline sample contains 756 securities, 30,819 firm-months, and 628,766 firm-days. Outcomes include quoted spread, effective spread, share depth, price improvement, trading volume, on/off NYSE liquidity measures, and NYSE market share.

Interpretation. These outcomes capture both posted liquidity and realized execution quality.

3 Models and empirical specifications

3.1 Treatment indicator

$$DMM_{i,t} = \mathbf{1}\{Vol_{i,t-1} - VT < 0\}, \quad VT = 1,000,000.$$

Interpretation. $DMM_{i,t}$ equals one when the security-month has the enhanced DMM contract.

3.2 Regression discontinuity specification

$$y_{i,t} = \alpha + \beta_0 DMM_{i,t} + \beta_1 [\ln Vol_{i,t-1} - \ln VT] + \beta_2 [\ln Vol_{i,t-1} - \ln VT] DMM_{i,t} + \gamma X_{i,t-1} + \eta_i + \nu_t + \varepsilon_{i,t}.$$

Interpretation. β_0 estimates the discontinuous change in market quality when the security crosses into the enhanced DMM contract.

3.3 Placebo and spillover specifications

The paper repeats the RD tests at fake thresholds, for Nasdaq securities, and for pre-contract NYSE securities. It also estimates the baseline specification separately for on-NYSE and off-NYSE outcomes.

Interpretation. The placebo tests support causality; the venue split identifies strategic complementarity.

4 Identification and empirical design

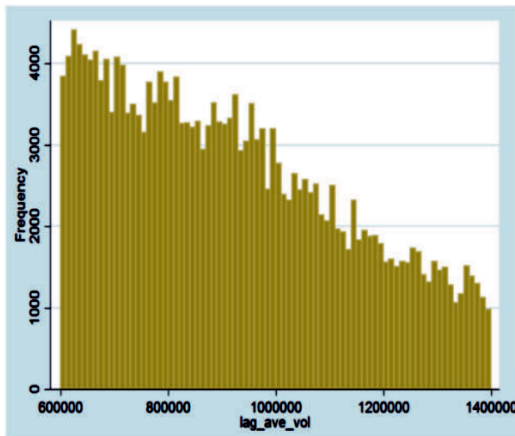
- The contract changes discretely at a pre-specified 1-million-share threshold.
- The same security can cross the threshold over time, allowing firm fixed effects.
- The density of the assignment variable is continuous at the cutoff.
- Placebo thresholds and placebo markets do not show the same pattern.
- A limitation is that obligations and rebates both change at the same cutoff, so their separate effects cannot be identified.

5 Tables and figures

5.1 Figure 1: Distribution of trading volume; Table 1: Sample summary statistics

Read: Figure 1 shows no obvious bunching around the threshold. Table 1 shows how narrower bandwidths improve balance between treated and control securities.

Economic message. The RD design is credible only if assignment is not precisely manipulated.



(a) Figure 1: Distribution of trading volume

Table 1
Sample summary statistics

Bandwidth (million shares)	None	0.5	0.1	0.05	0.025
Number of firms	756	738	606	512	399
Number of firm-months	30,819	16,228	3,196	1,585	726
Number of firm-days	628,766	336,850	66,409	33,019	16,074
Stronger DMM obligation	1 0	1 0	1 0	1 0	1 0
Consolidated average daily trading volume (1,000s)	537 1,845	730 1,222	949 1,049	974 1,025	988 1,012
Market capitalization (\$mil)	2,933 4,946	3,892 5,003	4,584 5,030	4,637 5,013	4,425 5,020
Share price (\$)	33.55 34.39	39.80 37.91	38.71 39.61	38.45 39.93	36.84 39.47
Turnover (%)	0.98 1.73	1.11 1.36	1.21 1.23	1.20 1.20	1.23 1.15
Return SD (%)	2.33 2.41	2.20 2.24	2.27 2.17	2.27 2.17	2.40 2.15
Quoted spread (basis points)	19.25 10.34	11.77 10.06	10.71 10.13	10.84 10.18	11.45 10.44
Effective spread (basis points)	14.97 8.47	9.29 8.10	8.59 8.07	8.65 8.10	9.16 8.34
Depth (shares)	1,331 2,659	1,369 1,956	1,642 1,606	1,724 1,491	1,814 1,509
Dollar depth (1,000\$)	23.61 41.11	28.32 35.81	31.64 33.25	31.88 32.45	32.03 32.26

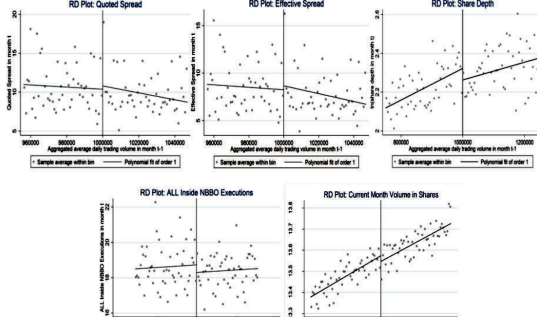
This table reports descriptive statistics for all NYSE-listed securities except those issued by financial and utility firms (bandwidth = none) that have at least one shift from consolidated average daily trading volume during a calendar month below 1 million shares to above 1 million shares or vice versa between September 2009 and December 2013. This table also reports descriptive statistics for the subsets of securities with bandwidths of 0.5, 0.1, 0.05, and 0.025 million shares. For each bandwidth, we report the number of firms, firm-months, and firm-days. All data are winsorized at the 0.5% level. Within each bandwidth, we report the descriptive statistics for the treatment group with stronger DMM obligation and the control group with weaker DMM obligation, respectively. Consolidated average daily trading volume is in thousands of shares; market capital is in million dollars; price of the security is in dollars; turnover and security daily return volatility are expressed as a percentage; the quoted spread and the effective spread are in basis points; share depth is in shares; and dollar depth is in thousands of dollar.

(b) Table 1: Sample summary statistics

5.2 Figure 2: Regression discontinuity plots; Table 2: RD tests for spreads, depth, and price improvement

Read: Figure 2 shows the visual discontinuities in spreads, depth, and inside-NBBO executions. Table 2 estimates quoted spread down 0.807 bps, effective spread down 0.736 bps, depth up, and price improvement up at the optimal bandwidth.

Economic message. The enhanced DMM contract improves both posted liquidity and execution quality.



(a) Figure 2: RD plots

Table 2
Regression discontinuity tests: Spreads, depth, and rate of price improvement

Column	Percentage quoted spread			Effective spread			ln(Share depth)			Percentage of execution within NBBO		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Bandwidth	> OB	> OB	< OB	> OB	> OB	< OB	> OB	< OB	< OB	> OB	> OB	< OB
DMM	0.0458 (0.14)	-0.807** (-2.32)	-1.187** (-2.22)	-0.0177 (-0.06)	-0.736** (-2.13)	-1.201** (-2.05)	-0.000101 (-0.03)	0.0127** (2.61)	0.0297** (2.03)	0.0839 (1.53)	0.326** (4.12)	0
ln(Vol)-ln(VT)	-4.607 (-0.97)	-11.88 (-1.34)	-85.48** (-2.37)	-5.320 (-1.35)	-13.60 (-1.53)	-88.79** (-2.86)	0.316** (5.78)	0.281** (5.64)	0.659** (4.71)	-1.188** (-2.28)	2.527** (3.82)	6
(ln(Vol)-ln(VT))* DMM	4.717 (0.75)	1.833 (0.15)	93.68* (0.88)	4.970 (0.96)	8.714 (0.81)	106.7** (2.55)	-0.108** (-2.55)	0.154** (2.56)	0.176 (0.85)	-1.451** (-2.11)	-1.187 (-0.60)	-5
Realized volatility	0.0232** (3.12)	0.0110** (2.09)	0.0101 (1.57)	0.0211** (3.49)	0.0107** (2.30)	0.00780 (1.32)	-0.0007** (-1.58)	-0.0005** (-4.39)	-0.0006** (-4.66)	0.0695** (8.11)	0.0770** (6.23)	0
Inv. price	25.84** (3.37)	41.97** (3.16)	53.61** (2.46)	24.78** (4.529)	40.70** (3.91)	45.09** (2.57)	0.625** (17.36)	0.861** (13.40)	1.561** (13.40)	-8.068** (-12.96)	-10.37** (-13.12)	-11
FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	55,834	27,970	13,431	55,834	27,970	13,431	160,780	82,518	38,905	99,598	50,376	2
Adj. R-sq.	.739	.714	.791	.707	.630	.689	.825	.846	.857	.583	.601	

This table reports the results of estimating the regression discontinuity specification (2) on four aspects of liquidity measures for securities that cross the trading volume threshold during the time period from September 2009 to December 2013. We estimate the following panel regression model:

(b) Table 2: RD tests

5.3 Table 3: RD tests for trading volume; Table 4: Placebo RD tests

Read: Table 3 shows higher trading volume under the enhanced DMM contract. Table 4 shows that fake thresholds, Nasdaq securities, and pre-contract NYSE data do not replicate the liquidity improvements.

Economic message. Placebo evidence supports the interpretation that the discontinuity is caused by the DMM contract.

Table 3
Regression discontinuity tests: Trading volume

Column	ln(Daily trading volume in shares)			ln(Daily trading volume in dollar)		
	(1)	(2)	(3)	(4)	(5)	(6)
Bandwidth	> OB	= OB	< OB	> OB	= OB	< OB
DMM	0.0125** (2.65)	0.0297*** (4.19)	0.0326*** (2.98)	0.0126** (2.31)	0.0288*** (3.50)	0.0361*** (3.11)
ln(Vol)-ln(VT)	0.615*** (22.48)	0.685*** (10.09)	1.054*** (5.40)	0.480*** (15.44)	0.621*** (8.06)	1.128*** (5.49)
(ln(Vol)-ln(VT))* DMM	-0.136*** (-4.36)	-0.137 (-1.72)	-0.789*** (-3.15)	0.0453 (1.30)	-0.175* (-1.96)	-1.201*** (-4.34)
Realized volatility	0.0002*** (2.42)	0.0004*** (3.62)	0.0003*** (2.52)	0.0001 (0.79)	0.0003*** (2.93)	0.0002 (1.69)
Inv. price	-0.113*** (-3.64)	-0.105*** (-1.85)	-0.0649 (-0.67)	-1.744*** (-38.35)	-1.814*** (-21.18)	-2.765*** (-21.84)
FEs	Yes	Yes	Yes	Yes	Yes	Yes
N	160,780	82,518	38,905	160,780	82,518	38,905
Adj. R-sq.	.214	.181	.206	.770	.777	.794

This table reports the results from the analysis using the regression discontinuity specification (2) with trading volume as the dependent variable on securities that cross the trading volume threshold during the time period from September 2009 to December 2013. We estimate the following panel regression model:

$$y_{i,t} = \alpha + \beta_0 DMM_{i,t} + \beta_1 (\ln(Vol_{i,t-1}) - \ln(VT)) + \beta_2 (\ln(Vol_{i,t-1}) - \ln(VT)) DMM_{i,t} + \gamma X_{i,t-1} + \eta_i + v_t + \varepsilon_{i,t},$$

where $y_{i,t}$ is the natural logarithm of daily trading volume in shares and in dollar for security i in calendar month t . $Vol_{i,t-1}$ is the consolidated average daily trading volume during the calendar month $t-1$ for security i in shares; $DMM_{i,t}$ is an indicator that equals 1 if a security is associated a optimal bandwidth (= OB), and a half of the optimal bandwidth (< OB). The optimal bandwidth is based on that of Calonico et al. (2017) and is

(a) Table 3: Trading volume

Table 4
Placebo regression discontinuity tests

	A. NYSE securities with a 0.5-million-share threshold (Sep. 2009-Dec. 2013)						B. NYSE securities with a 1.5-million-share threshold (Sep. 2009-Dec. 2013)					
	Quoted spread	Effective spread	ln(Share depth)	Inside NBBO	ln(Share volume)	ln(Dollar volume)	Quoted spread	Effective spread	ln(Share depth)	Inside NBBO	ln(Share volume)	ln(Dollar volume)
OB (mil.)	0.04	0.04	0.02	0.07	0.07	0.07	0.13	0.13	0.05	0.25	0.12	0.1
DMM	-0.053 (-0.09)	-0.045 (-0.09)	-0.021 (-0.53)	-0.022 (-0.32)	0.011 (0.55)	0.036 (1.34)	-0.203 (-0.61)	-0.164 (-0.57)	-0.009 (-0.19)	0.0201 (0.40)	-0.628 (-0.88)	-0.6
ln(Vol)-ln(VT)	-14.66 (-1.58)	-11.39 (-1.58)	0.164 (0.14)	0.724 (1.11)	0.847** (4.42)	1.137** (3.96)	0.845 (0.18)	0.521 (0.13)	-0.911 (-1.49)	0.834** (2.50)**	-0.152 (-0.40)	0
(ln(Vol)-ln(VT))* DMM	16.68 (1.14)	12.53 (1.11)	0.747 (0.48)	-4.06** (-4.72)	-0.40 (-1.64)	-0.68* (-1.54)	-2.472 (-0.45)	-1.342 (-0.20)	1.810 (0.84)	-2.50** (-4.65)	0.552 (1.16)	0
Realized volatility	0.012** (2.07)	0.012** (2.30)	-0.017 (-1.12)	0.06** (7.28)	0.0001 (0.69)	-0.0004 (-0.81)	0.014* (1.81)	0.012* (1.77)	-0.001 (-0.16)	0.025** (3.46)	-0.005 (-1.53)	-0
Inv. price	84.8** (4.14)	65.4** (4.09)	2.33** (5.87)	-5.92** (-9.51)	-0.145 (-0.98)	-1.05 (-1.61)	86.51** (10.09)	71.19** (9.83)	3.435** (4.44)	2.532** (6.62)	0.308 (1.07)	-3
FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	42,118	42,118	21,442	107,234	76,025	76,025	45,773	45,773	18,999	106,733	41,108	41

(b) Table 4: Placebo tests A-B

C. NASDAQ securities with a 1-million-shares threshold (Sep. 2009–Dec. 2013)

D. NYSE securities with a 1-million-shares threshold (Jul. 2005–Oct. 2008)

	Quoted spread	Effective spread	ln(Share depth)	Inside NBBO	ln(Share volume)	ln(Dollar volume)	Quoted spread	Effective spread	ln(Share depth)	Inside NBBO	ln(Share volume)	ln(Dollar volume)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
OB (mil.)	0.12	0.12	0.10	0.06	0.06	0.06	0.06	0.06	0.06	0.08	0.08	0.08
DMM	0.055 (0.08)	−0.048 (−0.08)	−0.0004 (−0.01)	0.598 (1.23)	0.024 (0.55)	0.029 (0.52)	0.055 (1.13)	0.449 (0.96)	−0.002 (−0.10)	0.088 (0.34)	0.0063 (0.29)	−0.0003 (−0.01)
ln(Vol)-ln(VT)	−0.562 (−0.08)	0.476 (0.08)	0.160 (0.29)	6.884 (0.62)	0.358 (0.41)	−0.589 (−0.49)	−9.320 (−1.05)	−2.063 (−0.26)	−0.578 (−1.15)	−2.225 (−0.52)	0.369 (1.04)	−0.0003 (−0.01)
(ln(Vol)-ln(VT))DMM	0.334 (0.03)	−5.016 (−0.55)	−0.053 (−0.08)	5.517 (0.38)	−0.247 (−0.21)	0.604 (0.37)	28.66* (1.83)	15.23 (1.05)	1.304* (1.86)	5.845 (1.01)	−0.0478 (−0.10)	−0.0003 (−0.01)
Realized volatility	0.0642*** (2.50)	0.063*** (2.73)	0.0005 (1.17)	−0.003 (−0.52)	−0.0003 (−0.59)	−0.0006 (−1.33)	0.118** (2.52)	0.100*** (2.29)	0.001** (2.44)	−0.003 (−0.58)	0.0002 (0.44)	−0.0003 (−0.01)
Inv. price	24.59*** (4.96)	22.41*** (4.62)	0.404** (2.56)	−0.621 (−0.37)	0.044 (0.48)	−1.274*** (−5.24)	82.67*** (2.84)	70.57*** (2.92)	1.750** (4.50)	−7.758* (−1.91)	−0.194 (−0.39)	−0.0003 (−0.01)
FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	57,348	57,348	47,295	28,254	29,346	29,346	46,623	46,623	46,623	60,992	61,622	23
Adj. R-sq.	.702	.524	.824	.691	.320	.816	.754	.623	.795	.696	.287	.754

This table reports the placebo test results of estimating the regression discontinuity specification (2) with the following panel regression model:

$$y_{i,t} = \alpha + \beta_0 DMM_{i,t} + \beta_1 (\ln(Vol_{i,t-1}) - \ln(VT)) + \beta_2 (\ln(Vol_{i,t-1}) - \ln(VT)) DMM_{i,t} + \gamma X_{i,t-1} + \eta_i + \nu_t + \epsilon_{i,t},$$

where $y_{i,t}$ is the daily percentage quoted spread, effective spread, the natural logarithm of share depth, percentage inside NBBO execution, and the natural logarithms of daily trading in shares and in dollar for security i in month t . $Vol_{i,t-1}$ is the consolidated average daily trading volume during calendar month $t-1$ for security i in million shares; $DMM_{i,t}$ is an indicator that equals 1 if $Vol_{i,t-1}$ is less than the threshold and 0 if $Vol_{i,t-1}$ is equal to or greater than the threshold; and $Vol_{i,t-1}$ is reduced by VT (the 1-million-shares threshold) to h

Table 4: Placebo tests C–D

5.4 Figure 3: On versus off NYSE RD plots; Table 5: On/off NYSE outcomes; Table 6: NYSE market shares

Read: Figure 3 and Table 5 show that off-NYSE price improvement and depth rise when the NYSE DMM contract is enhanced. Table 6 shows NYSE market shares decline.

Economic message. The improvements spill over to competing venues, supporting strategic complementarity rather than a purely mechanical DMM compliance effect.

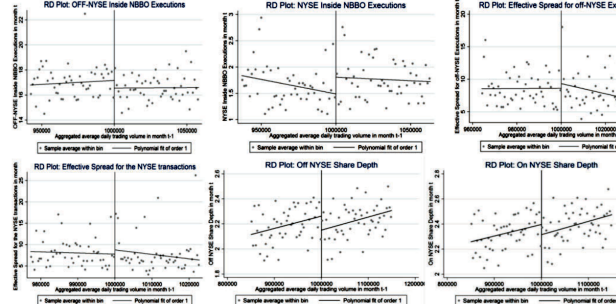


Figure 3: On/off NYSE RD plots

Table 5
Regression discontinuity tests for on versus off NYSE outcomes: Rate of price improvement, effective spread, and depth

	Percentage of execution within NBBO		Effective spread		ln(Share depth)	
	(1)	(2)	(3)	(4)	(5)	(6)
	Off-NYSE	On-NYSE	Off-NYSE	On-NYSE	Off-NYSE	On-NYSE
DMM	0.335*** (4.49)	-0.0126 (-0.56)	-0.768** (-2.18)	-0.636* (-1.86)	0.0128** (2.55)	0.0003 (0.06)
ln(Vol)-ln(VC)	2.844** (2.11)	-0.323 (-0.93)	-15.45 (-1.72)	-9.706 (-1.13)	0.266*** (5.16)	0.246*** (4.94)
(ln(Vol)-ln(VC))DMM	-0.756 (-0.41)	-0.581 (-1.15)	11.88 (1.07)	1.560 (0.15)	0.148** (2.41)	0.177 (2.89)
Realized volatility	0.0765*** (6.42)	0.001 (0.39)	0.0111** (2.25)	0.0104*** (2.94)	-0.0005*** (-8.04)	-0.0006** (-8.67)
Inv. price	-7.487*** (-10.74)	-2.836*** (-13.11)	41.28*** (4.06)	44.87*** (3.65)	0.860*** (13.31)	0.416*** (5.66)
FEs	Yes	Yes	Yes	Yes	Yes	Yes
N	50,376	50,376	27,970	27,970	82,466	82,466
Adj. R-sq.	.580	.654	.622	.844	.827	.799

This table reports the results from the analysis using the regression discontinuity specification (2) for on versus off NYSE activities on securities that cross the trading volume threshold during the time period from September 2009 to December 2013. We estimate the following panel regression model:

$$y_{i,t} = \alpha + \beta_0 DMM_{i,t} + \beta_1 (\ln(Vol_{i,t-1}) - \ln(VT)) + \beta_2 (\ln(Vol_{i,t-1}) - \ln(VT)) DMM_{i,t} + \gamma' X_{i,t-1} + \eta_i + \nu_t + \varepsilon_{i,t},$$

(a) Table 5: On/off NYSE outcomes

Table 6
Regression discontinuity tests for NYSE market shares

	NYSE market shares in number of transactions	NYSE market shares in total share volume	NYSE market shares in total dollar volume
	(1)	(2)	(3)
DMM	-0.278** (-2.55)	-0.295** (-2.18)	-0.295** (-2.18)
ln(Vol)-ln(VT)	-4.016** (-1.98)	-0.849 (-0.34)	-0.865 (-0.35)
(ln(Vol)-ln(VT))DMM	-3.116 (-1.13)	-5.326 (-1.59)	-5.306 (-1.58)
Realized volatility	-0.0008 (-0.60)	-0.006*** (-3.86)	-0.006*** (-3.86)
Inv. price	-2.279*** (-2.38)	-1.321 (-1.75)	-1.326 (-1.75)
FEs	Yes	Yes	Yes
N	38,905	38,905	38,905
Adj. R-sq.	.550	.497	.497

This table reports the results from the analysis using the regression discontinuity specification (2) with the NYSE market share as the dependent variable on securities that cross the trading volume threshold during the period from September 2009 to December 2013. We estimate the following panel regression model:

$$y_{i,t} = \alpha + \beta_0 DMM_{i,t} + \beta_1 (\ln(Vol_{i,t-1}) - \ln(VT)) + \beta_2 (\ln(Vol_{i,t-1}) - \ln(VT)) DMM_{i,t} + \gamma' X_{i,t-1} + \eta_i + \nu_t + \varepsilon_{i,t},$$

where $y_{i,t}$ is the NYSE market share in percentage, measured by number of trades, share volume, and dollar volume for security i in calendar month t ; $Vol_{i,t-1}$ is the consolidated average daily trading volume during

(b) Table 6: NYSE market shares

6 Short summary

- Enhanced DMM obligations at the NYSE improve liquidity near the 1-million-share threshold.
- Quoted and effective spreads fall, depth rises, and price improvement increases.
- Trading volume rises after the enhanced contract.
- Placebo tests rule out spurious volume-threshold effects.
- Off-NYSE liquidity improves even though the contract directly targets the NYSE DMM.
- NYSE market share falls, suggesting liquidity improvements are not just migration to the NYSE.
- The results support strategic complementarity in liquidity provision.

7 Conclusion

7.1 Core conclusion

Bessembinder, Hao, and Zheng show that enhanced NYSE DMM obligations improve market quality through narrower spreads, greater depth, and more frequent price improvement.

7.2 Mechanism

The mechanism is not only mechanical compliance. Off-NYSE markets also improve, suggesting that stronger DMM quoting makes liquidity provision by competitors safer or more profitable.

7.3 Market-design implications

Formal market-making contracts can be valuable even in fragmented electronic markets. They may stabilize liquidity supply and create amplified improvements through strategic complementarity.

7.4 Limitations

Obligation and rebate changes occur together at the cutoff, so the paper cannot separately identify their effects. The RD estimate is also local to securities near the 1-million-share threshold.

7.5 Takeaway

Liquidity provision is an equilibrium activity. A contract aimed at one designated market maker can change competitors' liquidity supply and improve market quality across venues.